

Prince Sultan University

Department of Computer & Information Sciences

IS241 – Database Management and Applications

#1067

Airline System

Prepared By:

Deema Abanmi-

Norah Bin

Daad Al Jaber -

Supervised By Dr. Amirah Alghanim

Table of contents

Introduction: 2

Business Rules: 2

ER Diagram: 3

Description of each entity: 4

Data Requirements: 4

Relational Model: 4

Data Dictionary; 4

Specification for search: 4

Specification for insert, delete and modify records: 4

Description of the organization :

The airline industry plays an important role in global transportation. Our proposed Airline Management System is designed for airlines that operate flights domestically and internationally, to streamline operations by managing flights, crew members, baggage handling, ticketing and passengers. The system ensures smooth and efficient booking and flight management. It benefits both airline employees and passengers, improving passenger experience by operating efficiently.

Purpose :

The purpose of the AMS is to automate and enhance airline management system operations, including flight scheduling, crew assignments, ticket reservations, passenger check-ins ,and baggage tracking. The system makes sure of smooth operating between different departments and provides real-time availability of important data. Improving the overall customer experience and satisfaction.

Objectives:

- **Airport and flight scheduling**: Retain accurate departure and arrival locations and timing for the flights and ensure valid arrangements are made with airports.

- **Crew management**: maintain valid records of pilot rank , flight hours, and emergency training levels for all flight attendants.

- **Flight operations management**: guarantee each flight is assigned to a specific plane and has a full crew team.

- **Efficient ticket handling**: allow passengers to book multiple tickets while making sure each ticket is linked to a specific passenger and flight.

- **Seating and baggage management**: Allocate tickets to specified seats , manage baggage owners and ensure it’s linked to a valid flight and passenger

- **Regulatory observation**: apply business rules such as accurate baggage tracking, required crew assignments and flight information and documentation.

Scope:

the airline management system handles multiple aspects such as:

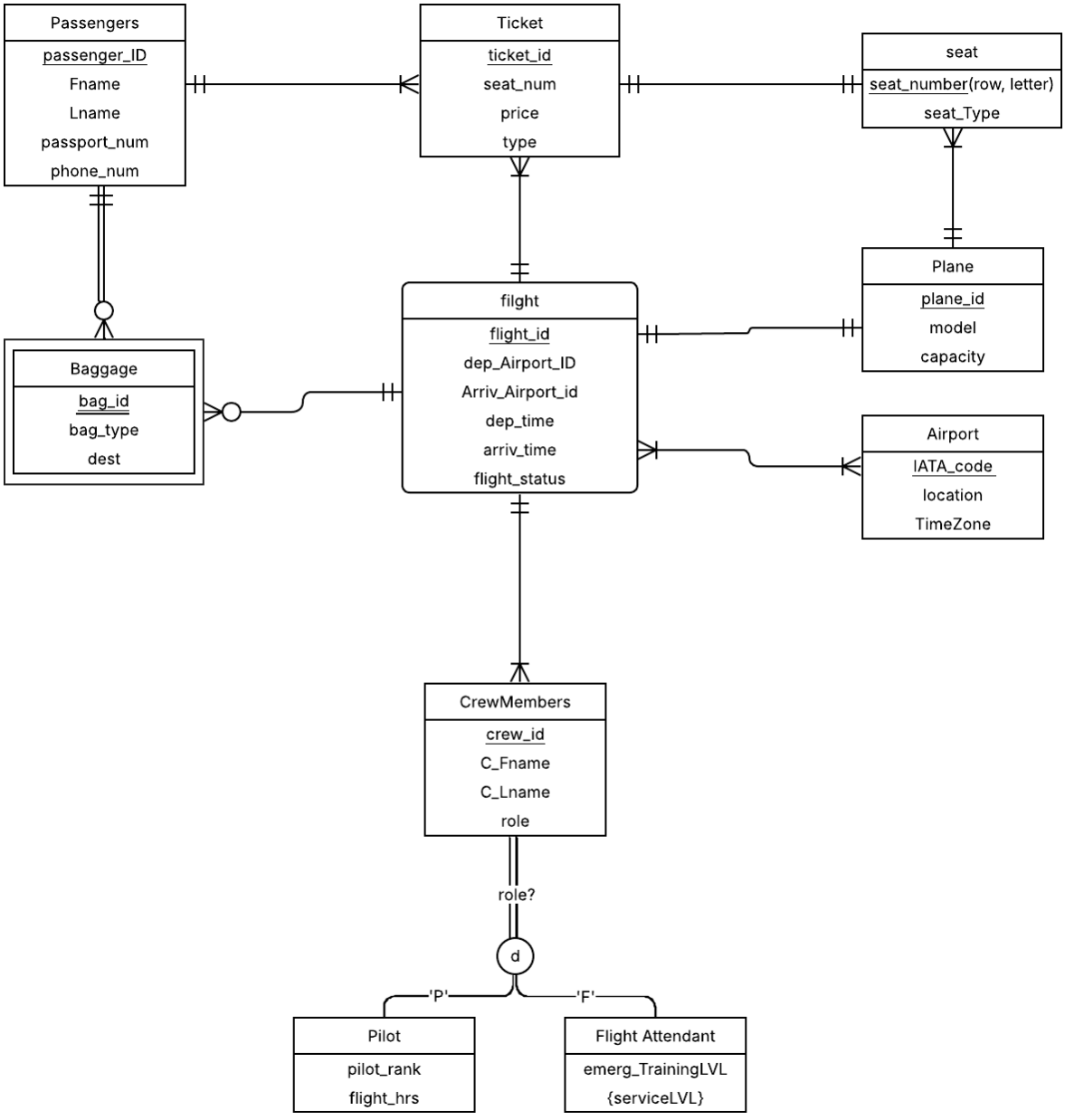
- **Passenger and ticket management**: managing the ticket reservation, seat, assignment, and passenger information.
- **baggage handling:** tracking baggage linked to passengers, and flights.
- **Flight scheduling and operation:** managing flight assignments, and aircraft deployment.
- **Crew management:** assigning pilots, and flight attendants based on their experience and qualifications.
- **Airport and route management:** managing the airport assignment, and guarantee that the flight has about the departure and arrival locations.
- **Data storage:** keeping records of flight ,passengers, crew members ,and operations for future reporting.

Business Rules:

1. A passenger can buy multiple tickets, but each ticket belongs to only one passenger.
2. A ticket is issued for one specific flight, and a flight can have multiple tickets.
3. A flight is operated by one plane, and a plane is operating one flight at a time.
4. A flight must have at least one CrewMember.
5. Crew members must be either a pilot or a flight attendant.
6. Each pilot must have a Pilot Rank (Captain, First Officer) and a record of Flight Hours to track experience.
7. Each flight attendant must have a designated Emergency Training Level (Basic, Advanced), a Uniform Type, and assigned service levels.
8. Each ticket must be assigned one seat, and a seat can only be assigned to one ticket per flight.
9. A passenger may check in multiple baggage items, but each baggage item belongs to only one passenger, bagages can not exist without a passenger.
10. A flight can carry no bagages or multiple bagages, but bagages is only carried on one flight.

11. A flight must have a valid departure and arrival airport, and an airport can have multiple flights assigned to it.

ER Diagram:

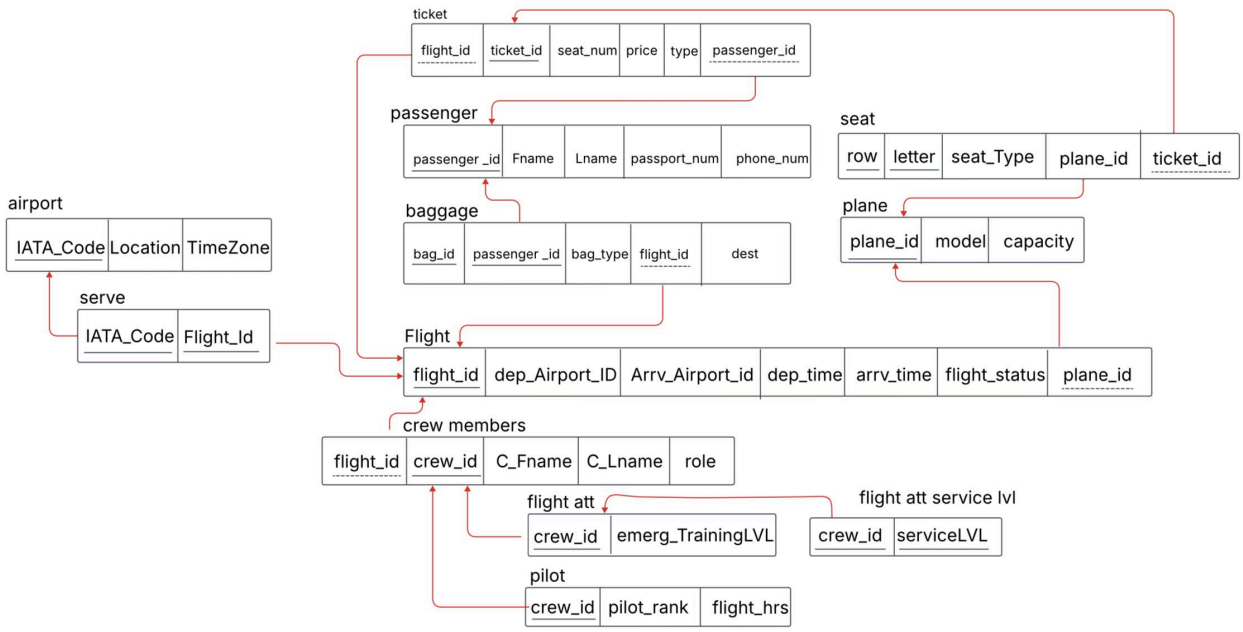


Entity description & Data requirements:

- Passengers: Represents the customers who buy tickets for flights.
- Attributes:
    - passenger_ID (Primary Key): Unique identifier for each passenger.
    - Fname: Passenger's first name.
    - Lname: Passenger's last name.
    - passport_num: Unique passport number.
    - phone_num: Passenger's contact number.
- Ticket: Represents the ticket purchased by a passenger for a flight.
- Attributes:
    - ticket_id (Primary Key): Unique identifier for each ticket.
    - seat_num: Seat assigned to the passenger.
    - price: price of the ticket.
    - type: Ticket type (e.g., economy, business, first class).
- Flight: Represents a scheduled flight from one airport to another.
- Attributes:
    - flight_id (Primary Key): Unique identifier for each flight.
    - dep_Airport_ID: The airport where the flight departs.
    - Arriv_Airport_id: The airport where the flight arrives.
    - dep_time: Scheduled departure time.
    - arriv_time: Scheduled arrival time.
    - flight_status: Current status of the flight (e.g., scheduled, delayed, canceled).
- Baggage: Represents luggage checked in by passengers.
- Attributes:
    - bag_ID (Primary Key):Unique identifier for each baggage item.
    - bag_type: Type of baggage (e.g., carry-on, checked-in).
    - dest: The final destination of the baggage.
- Seat: Represents a seat on a plane.
- Attributes:
    - seat_number (Primary Key): Unique composite identifier of the seat (row number and letter).
    - seat_Type: Type of seat (e.g., economy, business, first class).
- Plane: Represents the aircraft assigned to flights.
- Attributes:
    - plane_id (Primary Key): Unique identifier for each plane.
    - Model: Aircraft model (e.g., Boeing 747, Airbus A320).
    - Capacity: The maximum number of passengers the plane can allow.
- Airport: Represents the airports where flights take off and land.
- Attributes:
    - IATA_code (Primary Key): Standardized three-letter airport code (e.g., LAX, JFK).
    - location: Geographical location of the airport.
    - TimeZone: The time zone in which the airport operates.
- CrewMembers: Represents airline staff working on flights.

- Attributes:
    - crew_id (Primary Key): Unique identifier for each crew member.
    - C_Fname: Crew member's first name.
    - C_Lname: Crew member's last name.
    - role: Assigned role (pilot or flight attendant).
- Pilot: Subset of CrewMembers specifically trained to operate flights.
- Attributes:
    - pilot_rank: Rank of the pilot (e.g., Captain, First Officer).
    - flight_hrs: Number of hours the pilot has flown.
- Flight Attendant: Subset of CrewMembers responsible for passenger service and safety.
- Attributes:
    - emerg_TrainingLVL: Level of training for emergency situations.
    - serviceLVL:: Level of training for flight services. A flight attendant can be qualified for multiple service levels.

Relational Model:



Data Dictionary:

Table	column	Data type	Description	Constraints
passenger	passenger_id	INT	A unique ID for each passenger	Primary Key
	Fname	VARCHAR	The Passenger's first name	Not null
	Lname	VARCHAR	The Passenger's first name	Not null
	passport_num	VARCHAR	A unique passport number	Not Null
	phone_num	VARCHAR	Passenger’s contact number	Not Null
Ticket	ticket_id	INT	A unique ID for each ticket	Primary Key
	passenger_id	INT	References a specific passenger	Foreign key
	flight_id	INT	References a specific flight	Foreign key
	seat_num	VARCHAR	A seat assigned to a passenger	Not Null
	price	DECIMAL	The price of the ticket	Not Null
	type	VARCHAR	The ticket type (economy, first class)	Not Null
Flight	flight_id	INT	A unique ID for each flight	Primary key
	plane_id	INT	References a specific plane	Foreign key
	dep_Airport_ID	VARCHAR	Departure airport reference	Foreign key
	Arrv_Airport_id	VARCHAR	Arrival airport reference	Foreign Key
	dep_time	DATETIME	Date and time of flight departure	Not null
	arrv_time	DATETIME	Date and time of flight arrival	Not null

	flight_status	VARCHAR	Current status of the flight	Not Null
Plane	plane_id	INT	A Unique ID for each plane	Primary key
	model	VARCHAR	Description of the plane model	Not null
	capacity	INT	Maximum capacity of the plane	Not null
CrewMember	crew_id	INT	A unique ID for each crew member	Primary key
	C_Fname	VARCHAR	Crew member's first name	Not null
	C_Lname	VARCHAR	Crew member's last name	Not Null
	role	ENUM	Pilot or flight attendant (assigned role).	Not null
	flight_id	INT	References a specific flight	Foreign key
Flight Att	crew_id	INT	A unique ID for each flight attendant	Primary key
	emerg_TrainingLVL	VARCHAR	Emergency training level for attendants	Nullable
Flight Att Service LVL	crew_id	INT	A unique ID for each flight attendant	Primary key
	serviceLVL	VARCHAR	Service level for attendants	Nullable
Pilot	crew_id	INT	A unique ID for each pilot	Primary key
	pilot_rank	VARCHAR	Rank for pilots (Captain, etc..)	Nullable
	flight_hrs	INT	The flight hours for pilots	Nullable
Baggage	bag_id	INT	A unique ID for each baggage	Primary key
	passenger_id	INT	Belongs to a specific passenger	Foreign key
	bag_type	VARCHAR	Type of baggage	Not Null
	flight_id	INT	Belongs to a specific	Foreign key

			flight	
Seat	row	INT	A row number for the seat	Primary Key
	letter	VARCHAR	A letter designation of the seat	Primary Key
	seat_Type	VARCHAR	Type of seat (economy, business, first class)	Not Null
	plane_id	INT	References a specific plane	Foreign key
	ticket_id	INT	references a specific ticket	Foreign Key
Airport	IATA_Code	VARCHAR	A standard three letter airport code (RUH, DXB)	Primary Key
	location	VARCHAR	Location of the airport	Not null
	TimeZone	VARCHAR	Time zone in which the airport operates	Not Null

Specification for search 15 Queries:

1. **Find every flights leaving an airport:**
    - Input: IATA_Code (Departure)
    - Output: Flight ID, Departure Time, Arrival Time, Flight Status
2. **Find every flights arriving at an airport:**
    - Input: IATA_Code (Arrival)
    - Output: Flight ID, Departure Time, Arrival Time, Flight Status
3. **Locate all flights operated by a particular plane model.**
    - Input: Model
    - Output: Flight ID, Departure Time, Arrival Time
4. **Use the ticket ID to find passenger information:**
    - Input: Ticket ID
    - Output: Passenger ID, First Name, Last Name, Passport Number, Phone Number
5. **Search for all baggage linked to the passenger ID:**
    - Input: Passenger ID
    - Output: Baggage ID, Bag Type, Flight ID
6. **List every member of the crew on a particular flight:**
    - Input: Flight ID
    - Output: Crew ID, First Name, Last Name, Role
7. **Get the service level of the flight attendants on a particular flight:**
    - Input: Flight ID
    - Output: Crew ID, Service Level

8. **Get pilots' rank and flight hours on a particular flight:**
    - Input: Flight ID
    - Output: Crew ID, Pilot Rank, Flight Hours
9. **Find all flights managed by a particular crew member:**
    - Input: Crew ID
    - Output: Flight ID, Departure Time, Arrival Time
10. **Get all tickets for a particular flight:**
    - Input: Flight ID
    - Output: Ticket ID, Seat Number, Price, Type
11. **Search for seat details by using the ticket ID:**
    - Input: Ticket ID
    - Output: Row, Letter, Seat Type
12. **Find all flights scheduled within a particular date range:**
    - Input: Start Date, End Date
    - Output: Flight ID, Departure Time, Arrival Time, Flight Status
13. **List every planes and their capacity:**
    - No Input
    - Output: Plane ID, Model, Capacity
14. **Find all the flights handled by the pilots with a particular rank:**
    - Input: Pilot Rank
    - Output: Flight ID, Departure Time, Arrival Time
15. **Get all airports and their time zones:**
    - No Input
    - Output: IATA Code, Location, Time Zone

Specification for insert, delete and modify records:

Insert records:

- **Purpose**: adding new data entries to the database.

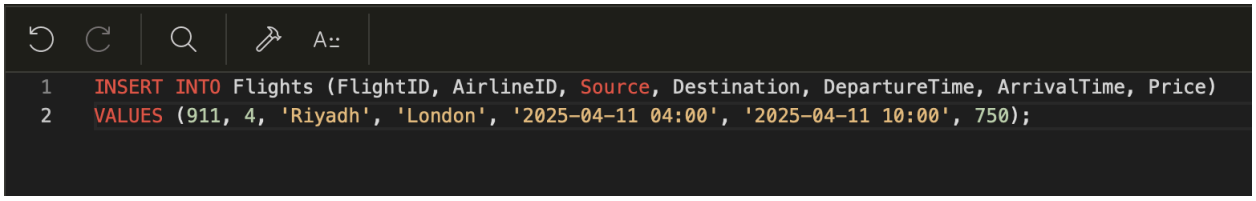
- Process:

- Start by collecting the required data from the user.
- Verify the accuracy and completeness of the data.
- Insert the validated data into the appropriate tables.

- Example: adding a new flight.

```
INSERT INTO Flights (FlightID, AirlineID, Source, Destination,
DepartureTime, ArrivalTime, Price)

VALUES (911, 4, 'Riyadh', 'London', '2025-04-11 04:00', '2025-04-11
10:00', 750);
```



```
INSERT INTO Flights (FlightID, AirlineID, Source, Destination, DepartureTime, ArrivalTime, Price)
VALUES (911, 4, 'Riyadh', 'London', '2025-04-11 04:00', '2025-04-11 10:00', 750);
```

Delete records:

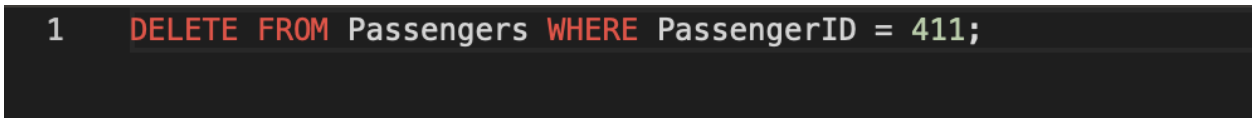
- **Purpose**: to remove from the database any incorrect or outdated data entries.

- Process:

- Identify the record to be destroyed using its primary key or a unique identifier.
- In order to prevent unintentional data loss, confirm deletion.
- Go ahead and delete the item.

- Example: delete passenger record.

```
DELETE FROM Passengers WHERE PassengerID = 411;
```



```
DELETE FROM Passengers WHERE PassengerID = 411;
```

Modify records:

- **Purpose**: to update and make changes to current data entries in order to maintain their accuracy.

- Process:

- Identify the record to be modified using its primary key or a unique identifier.
- Indicate which fields need to be updated along with their modified values.
- Validate the modified data before making any modifications.

- Example: updating flight details.

```
UPDATE Flights

SET Price = 800

WHERE FlightID = 911;
```

```sql
UPDATE Flights
SET Price = 800
WHERE FlightID = 911;
```